

Smart Energy Grid Monitoring System

Performance Analysis of TimescaleDB

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☰ Agenda

- System Overview
- Data Generation & Architecture
- Performance Experiments
- Key Findings
- Recommendations

- Real-time smart meter data via **MQTT (EMQX)**
- Time-series storage using **TimescaleDB**
- Python-based simulation and ingestion
- Performance-focused analytics

⚡ **Goal:** Analyze performance impact of chunking, compression, and aggregation

Smart Meters → MQTT Broker (EMQX) → Python Subscriber → PostgreSQL / TimescaleDB

- 500 smart meters
- Data every 5 minutes
- JSON-based message format

- 10-digit meter IDs
- Realistic daily load patterns
- Morning and evening peaks
- Night-time low consumption

Total Data Volume:  **4.2 million rows (2 weeks)**

- 3-hour chunks
- 1-day chunks
- 1-week chunks

 Cold cache testing after PostgreSQL restart

Q Performance Evaluation Queries

- Hourly average power
- Peak consumption detection
- Monthly energy per meter
- Full dataset scan

Execution time recorded for each chunk strategy

⚙️ Compression Impact

- Significant disk space reduction
- Improved analytical query performance
- Best results with time-ordered compression

↓ Compression Ratio up to **8x**

- 15-minute, hourly, daily aggregates
- Automatic refresh policies
- Faster dashboard queries

Query speed improved by **10–30x**



- Real-time readings (last hour)
- Daily comparison (today vs yesterday)
- Weekly trends
- Monthly usage by region

Includes live performance metrics panel



Key Findings

- 1 Chunk size strongly affects query latency
- 2 Compression drastically improves storage efficiency
- 3 Continuous aggregates outperform raw queries

✓ Recommendations

- Use **1-day chunks** for balanced workloads
- Enable compression for historical data
- Always use continuous aggregates for dashboards

TimescaleDB provides scalable, high-performance time-series analytics when properly configured.

 Ideal for smart energy monitoring systems

Thank You!